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CLIMATE REGULATING FEATURES FOR A REFRIGERATOR ICEBOX

Abstract

An ice making assembly mounted on a door of a refrigerator appliance includes an icebox mounted on the door and defining an ice making chamber and an icebox opening to the ice making chamber, an ice storage bin positioned below the icebox for storing ice, wherein an intake slot is defined between the icebox and the ice storage bin, a damper mounted over the icebox opening and being movable between an open position to permit a flow of air through the icebox opening and a closed position to prevent the flow of air through the icebox opening, a drive motor operably coupled to the damper, and a controller in operative communication with the drive motor for selectively pivoting the damper between the open position and the closed position.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to refrigerator appliances, and more particularly to door-mounted icemakers for refrigerator appliances.

BACKGROUND OF THE INVENTION

[0002] Refrigerator appliances generally include a cabinet that defines one or more chilled chambers for receipt of food articles for storage. Typically, one or more doors are rotatably hinged to the cabinet to permit selective access to food items stored in the chilled chamber. Further, refrigerator appliances commonly include ice making assemblies mounted within an icebox on one of the doors or in a freezer compartment. The ice is stored in a storage bin and is accessible from within the freezer chamber or may be discharged through a dispenser recess defined on a front of the refrigerator door.

[0003] It may be desirable to place craft icemakers on the freezer door for forming craft ice cubes (e.g., such as balls of ice greater than 2 inches in diameter), which are typically large cubes made by a conventional twist tray icemaker. Ideally, such an icemaker reliably produces high quality ice if the ambient temperature is maintained at a temperature elevated relative to the typical freezer compartment in which the icebox is located. Maintaining temperatures above this level may produce high-quality cubes with no cracks or surface bumps. In addition, these cubes may be easy to release from the icemaker mold. However, due to their positioning within the freezer compartment, door-mounted craft icemakers are typically maintained at too low a temperature, resulting in poor ice quality, poor harvest reliability, and consumer dissatisfaction. In addition, these compartments may prevent warm air from escaping, resulting in trapped humidity inside the compartment when water is added.

[0004] Accordingly, a refrigerator appliance with features for improved ice making would be desirable. More particularly, a refrigerator appliance with a door-mounted craft icemaker that is maintained at a temperature and humidity to produce high quality craft ice would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction is provided, including a cabinet defining a chilled chamber, a door being rotatably mounted to the cabinet to provide selective access to the chilled chamber, an icebox mounted on the door and defining an ice making chamber and an icebox opening to the ice making chamber, and a damper mounted over the icebox opening and being movable between an open position to permit a flow of air through the icebox opening and a closed position to prevent the flow of air through the icebox opening.

[0007] In another exemplary embodiment, an ice making assembly mounted on a door of a refrigerator appliance is provided. The ice making assembly includes an icebox mounted on the door and defining an ice making chamber and an icebox opening to the ice making chamber, an ice storage bin positioned below the icebox for storing ice, wherein an intake slot is defined between the icebox and the ice storage bin, a damper mounted over the icebox opening and being movable between an open position to permit a flow of air through the icebox opening and a closed position to prevent the flow of air through the icebox opening, a drive motor operably coupled to the damper, and a controller in operative communication with the drive motor for selectively pivoting the damper between the open position and the closed position.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. **1** provides a perspective view of a refrigerator appliance according to an exemplary embodiment of the present subject matter.

[0011] FIG. **2** provides a perspective view of the exemplary refrigerator appliance of FIG. **1**, with the doors of the fresh food chamber and freezer chamber shown in an open position.

[0012] FIG. **3** provides a side cross-sectional view of an icebox and ice making assembly for use with the exemplary refrigerator appliance of FIG. **1** according to an exemplary embodiment of the present subject matter.

[0013] FIG. **4** provides a front view of an icebox and ice making assembly for use with the exemplary refrigerator appliance of FIG. **1** according to an exemplary embodiment of the present subject matter.

[0014] FIG. **5** provides a side view of an icebox and ice making assembly for use with the exemplary refrigerator appliance of FIG. **1** according to an exemplary embodiment of the present subject matter.

[0015] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0017] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C. In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0018] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to

the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.
[0019] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.
[0020] FIG. **1** provides a perspective view of a refrigerator appliance **100** according to an exemplary embodiment of the present subject matter. Refrigerator appliance **100** includes a cabinet or housing **102** that extends between a top **104** and a bottom **106** along a vertical direction V, between a first side **108** and a second side **110** along a lateral direction L, and between a front side **112** and a rear side **114** along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another.
[0021] Housing **102** defines chilled chambers for receipt of food items for storage. In particular, housing **102** defines fresh food chamber **120** positioned at or adjacent second side **110** of housing **102** and a freezer chamber **122** arranged at or adjacent first side **108** of housing **102**. As such, refrigerator appliance **100** is generally referred to as a side-by-side refrigerator. It is recognized, however, that the benefits of the present disclosure apply to other types and styles of refrigerator appliances such as, e.g., a top mount refrigerator appliance, a bottom mount refrigerator appliance, or a single door refrigerator appliance. Consequently, the description set forth herein is for illustrative purposes only and is not intended to be limiting in any aspect to any particular refrigerator chamber configuration.
[0022] A refrigerator door **124** is rotatably hinged to an edge of housing **102** for selectively accessing fresh food chamber **120**. In addition, a freezer door **126** is rotatably hinged to an edge of housing **102** for selectively accessing freezer chamber **122**. Refrigerator door **124** and freezer door **126** are shown in the closed configuration in FIG. **1**. One skilled in the art will appreciate that other chamber and door configurations are possible and within the scope of the present invention.
[0023] A control panel **130** is provided for controlling the mode of operation. For example, control panel **130** includes one or more selector inputs **132**, such as knobs, buttons, touchscreen interfaces, etc., such as a water dispensing button and an ice-dispensing button, for selecting a desired mode of operation such as crushed or non-crushed ice. In addition, inputs **132** may be used to specify a fill volume or method of operating dispensing assembly **150**. In this regard, inputs **132** may be in communication with a processing device or controller **134**. Signals generated in controller **134** operate refrigerator appliance **100** and dispensing assembly **150** in response to selector inputs **132**. Additionally, a display **136**, such as an indicator light or a screen, may be provided on control panel **130**. Display **136** may be in communication with controller **134** and may display information in response to signals from controller **134**.
[0024] As used herein, “processing device” or “controller” may refer to one or more

microprocessors or semiconductor devices and is not restricted necessarily to a single element. The processing device can be programmed to operate refrigerator appliance **100** and dispensing assembly **150**. The processing device may include, or be associated with, one or more memory elements (e.g., non-transitory storage media). In some such embodiments, the memory elements include electrically erasable, programmable read only memory (EEPROM). Generally, the memory elements can store information accessible processing device, including instructions that can be executed by processing device. Optionally, the instructions can be software or any set of instructions and/or data that when executed by the processing device, cause the processing device to perform operations.

[0025] FIG. **2** provides a perspective view of refrigerator appliance **100** shown with refrigerator door **124** and freezer door **126** in the open position. As shown in FIG. **2**, various storage components are mounted within fresh food chamber **120** to facilitate storage of food items therein as will be understood by those skilled in the art. In particular, the storage components may include bins **140** and shelves **142**. Each of these storage components are configured for receipt of food items (e.g., beverages and/or solid food items) and may assist with organizing such food items. As illustrated, bins **140** may be mounted on refrigerator door **124** and freezer door **126** or may slide into a receiving space in fresh food chamber **120** or freezer chamber **122**. It should be appreciated that the illustrated storage components are used only for the purpose of explanation and that other storage components may be used and may have different sizes, shapes, and configurations.

[0026] Referring now generally to FIGS. **1** and **2**, a dispensing assembly **150** will be described according to exemplary embodiments of the present subject matter. Dispensing assembly **150** is generally configured for dispensing liquid water and/or ice. Although an exemplary dispensing assembly **150** is illustrated and described herein, it should be appreciated that variations and modifications may be made to dispensing assembly **150** while remaining within the present subject matter.

[0027] Dispensing assembly **150** and its various components may be positioned at least in part within a dispenser recess **152** defined on freezer door **126**. In this regard, dispenser recess **152** is defined on a front side **112** of refrigerator appliance **100** such that a user may operate dispensing assembly **150** without opening freezer door **126**. In addition, dispenser recess **152** is positioned at a predetermined elevation convenient for a user to access ice and enabling the user to access ice without the need to bendover. In the exemplary embodiment, dispenser recess **152** is positioned at a level that approximates the chest level of a user.

[0028] Dispensing assembly **150** includes an ice dispenser **154** including a discharging outlet **156** for discharging ice from dispensing assembly **150**. An actuating mechanism **158**, shown as a paddle, is mounted below discharging outlet **156** for operating ice or water dispenser **154**. In alternative exemplary embodiments, any suitable actuating mechanism may be used to operate ice dispenser **154**. For example, ice dispenser **154** can include a sensor (such as an ultrasonic sensor) or a button rather than the paddle. Discharging outlet **156** and actuating mechanism **158** are an external part of ice dispenser **154** and are mounted in dispenser recess **152**.

[0029] As shown in FIG. **2**, inside refrigerator appliance **100**, freezer door **126** may house one or more icemakers and ice storage bins that are configured forming and storing ice, respectively. In this regard, for example, freezer door **126** may include a first icebox **160** that includes an ice making chamber for housing ice making assemblies, storage mechanisms, and/or dispensing mechanisms. For example, first icebox **160** may be positioned proximate a top of freezer door **126** and may be designed for dispensing standard ice, e.g., through a front of freezer door **126** via dispensing assembly **150**.

[0030] In addition, according to an example embodiment of the present subject matter, freezer door **126** may also include a second icebox **162** that may house an ice making assembly **164** that operates independently of the icemaker in first icebox **160**. For example, ice making assembly **164** may be a craft icemaker for forming “craft ice” that is commonly large, clear cubes or spheres of ice for alcoholic or non-alcoholic drinks (e.g., as identified generally by reference numeral **166** in FIGS. **4** and **5**). A storage bin **168** may be positioned below icemaker for storing formed ice. A user may access this craft ice by opening freezer door **126** and accessing storage bin **168** directly.

[0031] Notably, the formation of craft ice may generally be improved if the ice formation temperature is elevated relative to conventional freezer temperatures. For example, conventional freezer temperatures hover around 0° F., whereas desired temperature for forming craft ice of high quality may be between about 15° F. and 20° F. Accordingly, aspects of the present subject matter generally directed to features of ice making assembly **164** and second icebox **162** that facilitate increased icebox temperatures and improved ice formation.

[0032] Notably, sealing off second icebox **162** may be desirable for maintaining suitable temperatures for forming craft ice. However, certain conditions may occur where warm, humid air inside second icebox **162** is unable to escape, e.g., when ice making assembly **164** is filled with water. If the humidity within second icebox **162** is unable to escape, frost may have a tendency to form on the inner walls of freezer door **126** and on ice making assembly **164**. This frost may build up over time and affect the operation of the icemaker when **164** and the ice formation process.

[0033] Referring now specifically to FIGS. **2** through **5**, ice making assembly **164** will be described in more detail according to example embodiments of the present subject matter. Specifically, according to the illustrated embodiment, the second icebox **162** may be mounted on freezer door **126** of refrigerator appliance **100** and may define an ice making chamber **200** and an icebox opening **202** to ice making chamber **200**. In general, ice making assembly **164** may be positioned within second icebox **162** where the temperature and humidity may be carefully regulated for forming high quality ice that may be easily removed from an ice mold **204** of ice making assembly **164**.

[0034] Specifically, according to the illustrated embodiment, second icebox **162** may be formed from a plurality of solid icebox walls **210**. In general, icebox walls **210** may include a front wall, a rear wall, two lateral sidewalls, and a top wall, each of which may be insulated. Icebox walls **210** generally define ice making chamber **200**. In addition, according to the illustrated embodiment, ice storage bin **168** may be positioned below second icebox **162** for storing ice **166**. According to the illustrated embodiment, an intake slot **212** may be defined between the ice storage bin **168** and a bottom of second icebox **162** along the vertical direction V. Intake slot **212** may generally be sized and configured for drawing in cool air (e.g., as identified generally by reference numeral **214**) to reduce the temperature within ice making chamber **200**.

[0035] In addition, icebox opening **202** may be defined in a top icebox wall **210** of second icebox **162**. In this manner, as warmer air rises, cool air **214** may be drawn in through intake slot **212** and may pass upward along the vertical direction V through ice making chamber **200** before exiting icebox opening **202**. According to example embodiments of the present subject matter, ice making assembly **164** may further include a damper **220** mounted over icebox opening **202** for regulating the flow of cool air **214**. In this regard, damper **220** may be movable between an open position to permit the flow of cool air **214** through icebox opening **202** and a closed position to prevent the flow of cool air **214** through icebox opening **202**. According to the illustrated embodiment, damper **220** includes a plurality of louvers that are pivotally mounted to top icebox wall **210**. However, it should be appreciated that according to alternative embodiments, any other suitable damper **220** or flow regulating device may be used while remaining within the scope of the present subject matter.

[0036] As illustrated, ice making assembly **164** may include a drive motor **222** that is operably coupled to damper **220**. Controller **134** may be in operative communication with drive motor **22** for selectively pivoting damper **220** between the open and closed position. Accordingly, controller **134** may regulate the position of damper **220** to control the temperature within ice making chamber **200**. In addition, as described in more detail below, damper **220** may be regulated to control the humidity within ice making chamber **200**.

[0037] According to an example embodiment, controller **134** may be configured to open damper

220 during the first portion of an ice making process and close damper **220** during a second, subsequent portion of the ice making process. In this regard, at the start of an ice making process, relatively warm water may be dispensed into ice mold **204**, thereby raising the humidity within ice making chamber **200**. By opening damper **220** at the start of the ice making process, the humidity within ice making chamber **200** may be quickly reduced, thereby preventing the formation of frost. In addition, the flow of cool air **214** may reduce the temperature within ice making chamber **200** to begin freezing the ice **166** within ice mold **204**. After the excessive humidity has been evacuated, it may be desirable to close damper **220**, e.g., to allow the temperature to increase and improve the ice formation process. By operating damper **220** as described, issues with excessive humidity (e.g., frost formation) may be reduced while the quality of ice cubes produced may improve.

[0038] According to still other embodiments, refrigerator appliance **100** may include a heating assembly **240** that is in thermal communication with second icebox **162** for selectively heating second icebox **162**. For example, heating assembly **240** may include a plurality of resistive heaters **242** that are positioned within second icebox **162** and freezer door **126**. Heating assembly **240** may be selectively operated in order to melt any collected frost, to regulate the temperature within second icebox **162**, etc.

[0039] Refrigerator appliance **100** may further include one or more temperature sensors **250** that are generally positioned for monitoring the temperatures of freezer chamber **122**, second icebox **162**, ice making chamber **200**, evaporator temperatures, etc. Specifically, according to the illustrated embodiment, temperature sensor **250** may be positioned within second icebox **162** for monitoring temperatures therein. Controller **134** may be in operative communication with temperature sensor **250** to measure temperatures, detect the presence of frost, etc. According to an example embodiment, one or more resistive heaters **242** may be positioned directly on damper **220**, e.g., to melt frost form thereon and to prevent damper **220** from locking up or binding. According to an example embodiment, controller **134** may be configured to turn on heating assembly **240** when damper **220** is in the closed position and an icebox temperature falls below a predetermined threshold.

[0040] As used herein, “temperature sensor” or the equivalent is intended to refer to any suitable type of temperature measuring system or device positioned at any suitable location for measuring the desired temperature. Thus, for example, temperature sensor **250** may each be any suitable type of temperature sensor, such as a thermistor, a thermocouple, a resistance temperature detector, a semiconductor-based integrated circuit temperature sensor, etc. In addition, temperature sensor **250** may be positioned at any suitable location and may output a signal, such as a voltage, to a controller that is proportional to and/or indicative of the temperature being measured. Although exemplary positioning of temperature sensors is described herein, it should be appreciated that refrigerator appliance **100** may include any other suitable number, type, and position of temperature, humidity, and/or other sensors according to alternative embodiments.

[0041] According to still other embodiments, controller **134** may be configured to determine that a dehumidification cycle is needed and periodically open damper **220** for a predetermined amount of time to purge humid air from within ice making chamber **200**. For example, controller **134** may be in operative communication with a humidity sensor **252** for determining when the humidity within ice making chamber **200** exceeds a predetermined threshold humidity. By contrast, determining that a dehumidification cycle is needed may be based on the elapsed time since commencing an ice making process. For example, damper **220** may be regulated based on how recently water has been dispensed into ice mold **204**. It should be appreciated that other means for determining that a dehumidification cycle is needed may be used while remaining within the scope of the present subject matter.

[0042] According to alternative embodiments, determining that a dehumidification cycle is needed may be based on elapsed time since a prior dehumidification cycle, an indication of frost buildup, or a number of icemaking cycles performed since the prior dehumidification cycle. According to still other embodiments, determining that the dehumidification cycle is needed may be based on a measured icebox temperature or humidity, a measured freezer temperature or humidity, and/or a measured freezer evaporator temperature. Similarly, controller **134** may be programmed to determine when frost has formed and may operate heating assembly **240** accordingly to adjust the temperature within second icebox **162**. Controller **134** may also use heating assembly **240** to regulate temperature to the desired ice formation temperature.

[0043] As explained herein, aspects of the present subject matter are generally directed to a frost removal damper for an ice maker. To make better quality ice cubes, the temperature above an ice mold body should be maintained higher than normal freezer compartment temperature but less than freezing temperature (15-20° F.). When the ice maker needs warmer temperature on top of ice mold body, the ice maker may preferably be closed above the ice mold body to hold warmer air efficiently. At the same time, an air inlet and outlet area are provided under the ice mold body to allow cold air to come into the icebox. For defrosting and better air circulation, the ice maker may include a damper (e.g., a shutter), and when the ice maker does not need warmer temperatures, the damper may be opened. This ice maker may be used to produce good quality “craft ice” (e.g., cocktail ice) which is bigger than normal ice cubes. The ice maker may include a twist tray system (e.g., a motor/gear/control box, an ice tray, a thermistor, etc.) for good quality and a damper assembly (e.g., a damper, a motor/gear, a damper heater, etc.). The damper may be open for a quick ice making mode and for the first several hours of ice formation, and the damper may be closed for the last portion of ice formation. The damper heater may be turned on only when the damper is closed and icebox temperature is cold.

[0044] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, comprising: a cabinet defining a chilled chamber; a door being rotatably mounted to the cabinet to provide selective access to the chilled chamber; an icebox mounted on the door and defining an ice making chamber and an icebox opening to the ice making chamber; and a damper mounted over the icebox opening and being movable between an open position to permit a flow of air through the icebox opening and a closed position to prevent the flow of air through the icebox opening.

2. The refrigerator appliance of claim 1, further comprising: an ice storage bin positioned below the icebox for storing ice, wherein an intake slot is defined between the icebox and the ice storage bin.

3. The refrigerator appliance of claim 1, wherein the icebox opening is defined in a top wall of the icebox.

4. The refrigerator appliance of claim 1, further comprising: a drive motor operably coupled to the damper; and a controller in operative communication with the drive motor for selectively pivoting the damper between the open position and the closed position.

5. The refrigerator appliance of claim 1, further comprising a controller configured to open the damper during a first portion of an ice making process and close the damper during a second portion of the ice making process.

6. The refrigerator appliance of claim 1, further comprising a controller configured to: determine that a dehumidification cycle is needed; and open the damper for a predetermined amount of time to purge humid air from the ice making chamber.

7. The refrigerator appliance of claim 6, wherein the determination that the dehumidification cycle is needed is based on an elapsed time since commencing an ice making process.

8. The refrigerator appliance of claim 1, further comprising: a heating assembly in thermal communication with the icebox for selectively heating the icebox.

9. The refrigerator appliance of claim 8, wherein the heating assembly comprises a resistive heater.

10. The refrigerator appliance of claim 8, further comprising a controller configured to: turn on the heating assembly when the damper is in the closed position and an icebox temperature falls below a predetermined threshold.

11. The refrigerator appliance of claim 1, further comprising: a temperature sensor positioned in thermal communication with the icebox.

12. The refrigerator appliance of claim 1, further comprising: a craft icemaker positioned within the icebox.

13. The refrigerator appliance of claim 1, wherein the chilled chamber is a freezer compartment of the refrigerator appliance.

14. The refrigerator appliance of claim 1, wherein the refrigerator appliance is a side-by-side refrigerator appliance.

15. An ice making assembly mounted on a door of a refrigerator appliance, the ice making assembly comprising: an icebox mounted on the door and defining an ice making chamber and an icebox opening to the ice making chamber; an ice storage bin positioned below the icebox for storing ice, wherein an intake slot is defined between the icebox and the ice storage bin; a damper mounted over the icebox opening and being movable between an open position to permit a flow of air through the icebox opening and a closed position to prevent the flow of air through the icebox opening; a drive motor operably coupled to the damper; and a controller in operative communication with the drive motor for selectively pivoting the damper between the open position and the closed position.

16. The ice making assembly of claim 15, wherein the icebox opening is defined in a top wall of the icebox.

17. The ice making assembly of claim 15, wherein the controller is configured to open the damper during a first portion of an ice making process and close the damper during a second portion of the ice making process.

18. The ice making assembly of claim 15, wherein the controller is configured to: determine that a dehumidification cycle is needed; and open the damper for a predetermined amount of time to purge humid air from the ice making chamber.

19. The ice making assembly of claim 18, wherein the determination that the dehumidification cycle is needed is based on an elapsed time since commencing an ice making process.

20. The ice making assembly of claim 15, further comprising: a heating assembly in thermal communication with the icebox for selectively heating the icebox, wherein the controller is configured to turn on the heating assembly when the damper is in the closed position and an icebox temperature falls below a predetermined threshold.